UNIT - V

Industrial Applications of NDE

1. Span of NDE Activities

* NDE is used in **every stage of the life of a product** :

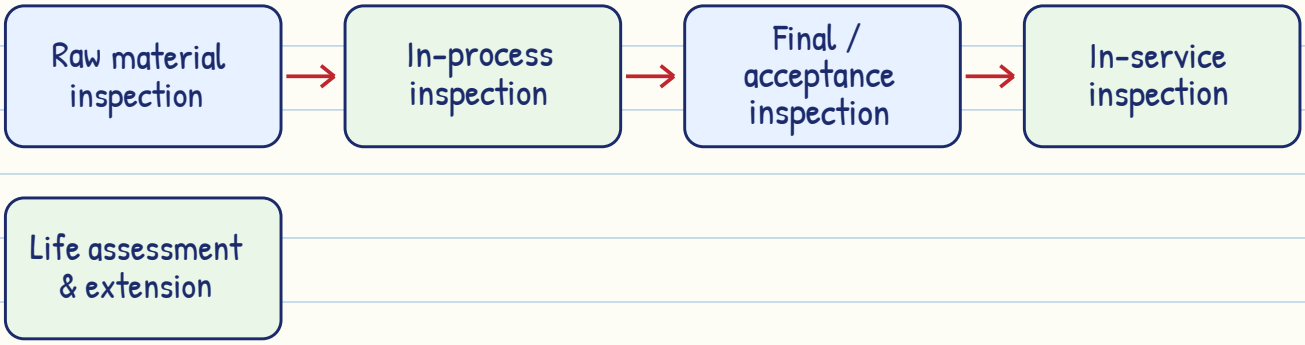


Fig: NDE through the life cycle

* **Raw material stage:** check ingots, plates, bars, forgings and castings for laminations, inclusions, cracks.
* **Manufacturing stage:** check welds, machined parts, heat treatment, assembly – feedback for process control.
* **Pre-service inspection (PSI)** – the baseline record before the plant starts.
* **In-service inspection (ISI)** – periodic inspection, to find fatigue cracks, corrosion, creep, erosion, stress corrosion cracking.
* **Condition monitoring and remaining life assessment.**
* **Failure analysis** after accidents.

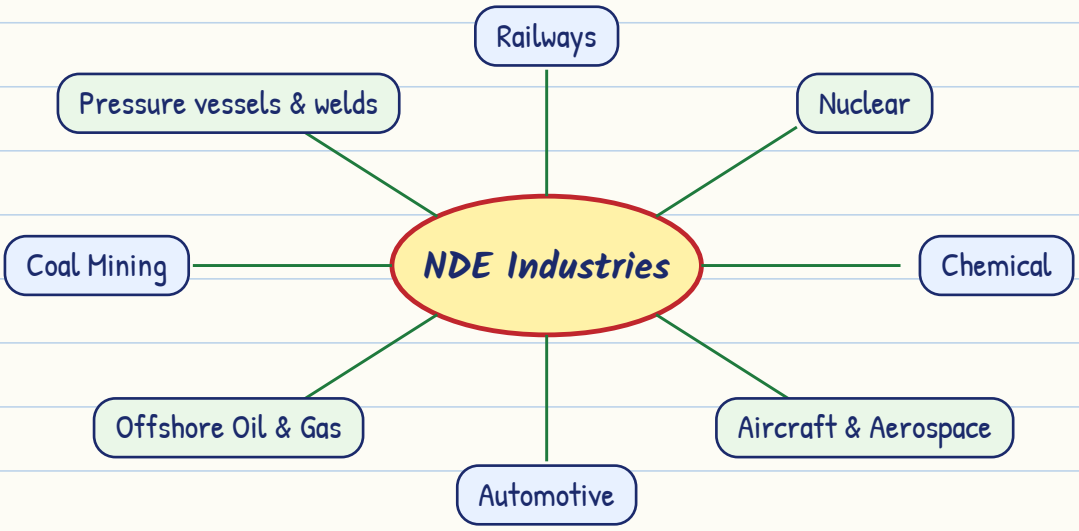


Fig: Span of NDE applications

Choice of NDE Method

Need	Preferred method
Surface cracks in ferromagnetic steel	MPT (with LPT as alternate)
Surface cracks in non-magnetic / non-porous	LPT, ET
Internal flaws in welds	RT, UT (TOFD, phased array)
Thickness and corrosion mapping	UT thickness gauge, RT profile
Tube inspection (heat exchanger)	Eddy current, remote field
Composite delamination, honeycomb	UT C-scan, thermography, shearography
Leak detection	Helium / bubble / acoustic / pressure decay
Hot spots, insulation, electrical	IR thermography
Fatigue crack growth monitoring	Acoustic emission

2. Railways

* Railways carry heavy loads at high speeds, so ***failure of rail, wheel or axle can cause derailment*** .
* ***Rail inspection: ultrasonic rail flaw detection*** using trains with probes at 0, 37 and 70 degree angles to find head checks, transverse defects (kidney flaws), ***detail fractures*** , web cracks and weld defects. Hand-pushed trolleys also used.

* **Eddy current testing** of rail head for **surface fatigue cracks (squats, gauge corner cracks)** and **magnetic flux leakage** for rails.
* **Rail weld inspection:** thermite / flash butt welds - UT and magnetic particle testing; radiography in some cases.
* **Wheels and axles: UT of axle (hollow and solid, from end face with angle probes)** to find fatigue cracks at press-fit seats; MPT after overhaul; **wheel tread ECT / UT** for cracks and shatter rim; wheel flat detection by **acoustic** or laser systems.
* **Bogie frames, couplers, springs, bolsters and castings** - MPT and LPT in workshops.
* **Bridges and tracks:** visual, UT of welded steel girders, rebound hammer and UPV on concrete sleepers, ground penetrating radar for ballast.
* **Hot box detectors** use passive **infrared sensors** to measure axle-box temperatures of passing trains.
* **Examples of defects:** rolling contact fatigue, corrugation, wheel burn, shelling.

3. Nuclear Industry

* Needs **the highest reliability and safety**; failure may cause radioactive release - every component is inspected by **strict codes (ASME Sec. XI, RCC-MR, AERB)** both before and during service.
* **Components:** reactor pressure vessel, steam generator tubes, primary coolant piping, fuel cladding (zircaloy tube), fuel pins, pressure tubes (PHWR), calandria, nozzle welds, turbines.
* **Methods used:**
  - **UT** - automated remote immersion scanning of reactor pressure vessel welds and nozzle to shell joints, **phased array and TOFD** on piping.
  - **Eddy current** - steam generator tubes (bobbin coil, rotating probes, array probes) to find wall thinning, pitting, fretting, stress corrosion cracks; **remote field ECT**.
  - **Radiography** - welds of piping and vessels, including high energy sources; **neutron radiography** for fuel and explosive devices.

- LPT / MPT for surface of components, leak testing (helium mass spectrometer) for containment and pipes.
- Visual inspection with remote cameras and underwater robots , acoustic emission for leak and crack monitoring.
- Gamma scanning, X-ray CT for fuel pellets and pins , dimension gauging of pressure tubes (ultrasonic wall thickness, diameter, sag).

* Special problems: high radiation field, limited access - need remote handling manipulators, robotic crawlers, shielding and dose control (ALARA) .
* Materials issues: irradiation embrittlement, stress corrosion cracking, hydrogen embrittlement (hydride blisters in zircaloy), thermal fatigue.
* Non-nuclear industries also use the same methods, but with lower requirements.

4. Non-nuclear and Chemical Industries

4.1 Non-nuclear (Thermal power, steel, general engineering)

* Boiler and turbine: boiler tubes - UT thickness, EMAT, RT of welds, ECT for tube thinning , creep and fatigue cracking; MPT of turbine blades, rotors, discs and shafts ; UT of rotor bores and forged shaft .
* Power plant: steam pipes (creep damage by replica metallography), headers, valve bodies, generator retaining rings (ECT), thermography of electrical boards , transformers and insulators.
* Steel plants: UT of rolled plates and bars for laminations, ET and MPT on billets, rolls; infrared of furnace refractory .
* Foundry and forge: RT, UT and MPT of castings and forgings.
* Bridges, buildings and civil structures: rebound hammer, ultrasonic pulse velocity (UPV), ground penetrating radar, half-cell potential for rebar corrosion, core test, MT on steel bridge weld.
* Wind turbines: blade UT, thermography, acoustic emission, tower weld MPT.

4.2 Chemical, Fertilizer and Petrochemical Industries

* Hazardous fluids at high pressure and temperature and corrosive environment -> leaks and failure cause fire, explosions and toxic release .

* **Equipment:** reactors, columns, heat exchangers, storage tanks and spheres, pipelines, furnaces, boilers, pumps, valves.
* **Typical damage:** general and **pitting corrosion, corrosion under insulation (CUI), erosion, stress corrosion cracking (SCC), hydrogen induced cracking (HIC), creep, fatigue, thermal fatigue, fouling** .
* **Methods:**
  - **UT thickness gauging and corrosion mapping (automated scanners, B/C-scan)** for tank floors and pipes.
  - **Radiography (including profile and real-time)** for CUI and weld inspection of pipe lines.
  - **Eddy current** on heat exchanger and condenser tubes, **IRIS (Internal Rotary Inspection System)** and **remote field ET** for ferromagnetic tubes.
  - **Magnetic flux leakage (MFL)** for tank floor and pipeline; **Guided wave UT** (long range) to check long pipe runs and under supports without removal of insulation.
  - **Wet fluorescent MPT** for SCC, **LPT** for stainless steel and non-magnetic alloys.
  - **Infrared thermography** - furnace tube temperature, refractory lining, steam trap, insulation loss.
  - **Acoustic emission** - online monitoring of pressure vessels and spheres, leak detection of valves.
  - **Risk based inspection (RBI)** decides the schedule and method according to risk.
* Inspection is done during **shutdown / turnaround** or on-stream.

5. Aircraft and Aerospace Industries

* **Safety critical** with high loads, fatigue and weight limits; **fail-safe and damage tolerance design** depend on NDE; inspection by regulations (DGCA, FAA, EASA) at specified flight-hour or cycle intervals (A, B, C, D checks).
* **Structures:** skin, fuselage lap joints, wing spars and ribs, landing gear, engine mounts, fasteners and rivet holes, control surfaces, bulkheads, propeller blades, helicopter rotor blades.

* **Typical defects:** fatigue cracks at fastener holes, corrosion, stress corrosion cracking, disbonds and delaminations, impact damage in composites, fretting, water in honeycomb.
* **Methods:**
  - **Eddy current** (surface probes, sliding probe for bolt holes, low frequency for sub-surface cracks under layers, conductivity for heat damage) - most used in-service method on **aluminium skin and engine disks** .
  - **Ultrasonic:** thick sections, forgings, bond testers (resonance, mechanical impedance analysis), **C-scan of composites** , phased array on wing, **Lamb waves** .
  - **Radiography:** castings, honeycomb water ingress, inaccessible regions; **real-time and digital radiography, CT** for turbine blades and cores.
  - **LPT (fluorescent)** on engine components, **MPT** on steel landing gear and bolts.
  - **Thermography** - delamination and disbond in composite panels, honeycomb, **space shuttle tiles** ; **shearography** for composite bond.
  - **Acoustic emission** for pressure tests and structural health monitoring; **tap test** for quick checking of the bonded skin.
  - **Borescope / video endoscope** for the inside of the engine and tanks.
* **Aerospace launch vehicles:** motor cases, cryogenic tank welds (RT, UT, helium leak test), solid propellant bond lines, nozzles; satellite structures.
* **Structural health monitoring (SHM)** with embedded sensors (fibre Bragg grating, piezo patches) for **real time** monitoring.

6. Automotive Industries

* Mass production and cost sensitive; need **fast, automatic, in-line** NDE on safety critical parts.
* **Engine: crank shaft, connecting rod, cam shaft, piston pin, gears** - **MPT** (automatic wet fluorescent benches) to find grinding cracks, forging laps; **ECT** for hardness, case depth, heat treatment, sorting of mixed parts.
* **Castings:** cylinder block and head, wheels (aluminium) - **radiography (real-time / digital X-ray, CT)** for porosity and shrinkage; **UT** for brake discs; LPT for leaks

and cracks.

* **Welds:** body in white **spot weld quality** by ultrasonic method (phased array), **laser weld** monitoring by IR, tube and pipes by ECT in production.
* **Brake and tyre: shearography / holography** of tyre sidewall and retreads, UT for brake lining bond, **eddy current for wheel rims** .
* **Fuel tanks, pipes, airbag inflators, radiators, AC lines - leak testing** (helium, pressure decay).
* **Adhesive joints** and composite bodies - ultrasonic, thermography.
* **Electronics and sensors** - X-ray for solder joints (BGA), IR thermography for hot spots in ECUs and battery cells (EV).
* **Final inspection:** vibration / acoustic noise tests (NVH), **eddy current sorting** of heat treated components.
* **Benefits:** zero defect supply, reduced warranty claims, recall prevention.

7. Offshore Gas and Petroleum Projects

* **Harsh environment:** seawater, waves and storms, fatigue, corrosion, marine growth, **remote access** and hazardous area; high cost of failure and pollution.
* **Structures:** jackets and platforms (steel tubular joints), risers, subsea pipelines, FPSO, drilling rigs, wellheads, pressure vessels and separators, crane, mooring chains, flexible risers, storage tanks.
* **Fabrication stage: RT, UT, MPT, LPT of welds** , mostly **AUT (automated UT), TOFD, phased array** for pipeline girth welds as per API 1104 / DNV; **alloy verification (PMI)** .
* **In-service inspection: underwater inspection by divers or ROV (remotely operated vehicle):**
  - **Visual (CCTV) and marine growth cleaning; flooded member detection (gamma or UT);**
  - **Alternating current field measurement (ACFM)** to find and size fatigue cracks at **tubular joint welds** even through coatings, without cleaning;
  - **Underwater MPT and eddy current; UT thickness** and corrosion mapping on risers, **cathodic protection (CP) potential survey** .

* **Pipeline pigging:** *intelligent pigs (in-line inspection tool)* run inside the pipeline using *magnetic flux leakage (MFL)* or *ultrasonic* sensors for finding corrosion, dents, cracks and wall loss over hundreds of kilometres.
* **Tanks and spheres:** floor corrosion by MFL, *acoustic emission* ; *guided wave* for risers and insulated lines.
* **Drilling equipment:** drill pipe, drill collars, BOP, wellheads - EMI (electromagnetic inspection), UT, MPT.
* **Leak detection:** acoustic, gas detectors, optical gas imaging with IR cameras.
* Standards: API 510, 570, 653, 1104, DNV-OS, NORSOK, ASME B31.3 and B31.8.

8. Coal Mining Industry

* Heavy-duty machinery in *dusty, wet, explosive* (methane and coal dust) atmosphere; *safety of miners* depends on integrity of equipment.
* **Hoisting equipment:** *wire ropes of winders and haulage* - the most important - inspected by *magnetic flux leakage (MFL) and magnetic rope testers* to find broken wires (loss of metallic area, LMA) and corrosion; *winding drums, shafts, sheaves, suspension gear* - MPT, UT; *cage and skip hangers, chains, hooks* - MPT.
* **Mining machines:** draglines, shovels, dumpers, conveyors, shearers, continuous miner - *UT and MPT of axles, shafts, gear, pins, welds in booms and buckets, bearing housings* ; *ECT* of hydraulic components.
* *Dump truck frames* and *rock bolts* - UT, *guided wave on rock bolts / anchor* .
* **Roof and rock mass:** *acoustic emission, microseismic monitoring* and *ground penetrating radar* for roof falls, strata monitoring, detection of voids, abandoned workings and water bodies.
* **Conveyor belts:** steel cord belt - X-ray or magnetic induction to find broken cords, joint failure; IR thermography of idlers and drive motors (fire prevention).
* **Pressure equipment:** boilers, compressors, air receivers - RT, UT.
* *Explosion proof and intrinsically safe NDT instruments* are required underground.

* **Thermography** to detect hot spots in coal stockpile (spontaneous heating), fires in coal seams.
* **Challenges:** dirt, grease, wet surfaces, dust, hazardous locations, limited time.

9. NDE of Pressure Vessels

* **Pressure vessels** (boilers, reactors, storage spheres, LPG bullets, air receivers, columns) hold fluids at **pressure different from the atmosphere** ; failure may be **catastrophic (explosion)** .
* Codes: **ASME Sec. VIII Div. 1 / 2, Sec. I (boilers), Sec. V (NDE methods), IS 2825, PD 5500, EN 13445** .
* **Parts:** shell, dished ends (heads), nozzles and manway, flanges, skirt / saddle, internals, welds (longitudinal, circumferential, nozzle to shell).
* **During fabrication:**
  - **Plates and forgings:** UT (laminations) as per ASTM A435 / A578; **PMI** for alloy.
  - **Weld edges and joints: MPT / LPT** for root and back chipping, **RT (spot or 100%, depending on joint efficiency 0.7 / 0.85 / 1.0)** and UT; **TOFD / PAUT** in place of RT for thick walls.
  - **Nozzle welds, attachment welds:** MPT or LPT, UT.
  - **Heat treatment check:** hardness test, **post weld heat treatment (PWHT)** records.
  - **Final hydrotest / pneumatic test** with **acoustic emission** monitoring.
* **In-service inspection:**
  - **External and internal visual,** UT thickness survey of shell and heads for corrosion allowance and remaining life; **wet fluorescent MPT** of welds (SCC, fatigue); **LPT on stainless steel; RT / UT / TOFD** for new crack growth; **hydrogen blistering and HIC** by UT shear wave (angle probes); **creep** by replica metallography and hardness; **thermography** for refractory; **acoustic emission** for on-line monitoring.
* **Inspection intervals** by **Risk Based Inspection (RBI)** and fitness for service (API 579).

* *Typical flaws:* weld cracks, lack of fusion, slag, porosity, corrosion pits, laminations, blisters, bulging.

10. NDE of Castings

* *Casting* = metal poured in a mould; defects occur in solidification and cooling.
* *Defects: blowholes and gas porosity, shrinkage cavity and porosity, hot tears and cold cracks, cold shuts, misruns, sand / slag inclusion, segregation, core shift, mismatch, surface roughness.*
* *Special problems of NDT on castings: coarse grain, rough surface, complex shape, varying thickness, scattering of ultrasound* (cast iron, steel, stainless).
* *Methods:*
  - *Visual inspection* and dimensional check - first step.
  - *Radiography* - the *best for volumetric flaws* (shrinkage, gas porosity, inclusion, hot tears); *film, real-time and CT for aluminium and titanium casting* ; graded by *ASTM E446 / E186 / E280 reference radiographs* .
  - *UT* - low frequency (0.5 - 2 MHz), large crystal, dual crystal probe, *also graded for porosity by back wall attenuation* ; good for thick steel castings.
  - *MPT* (steel, iron castings) for surface and near surface cracks, hot tears; *LPT* (aluminium, brass, stainless) for surface-opening flaws; dye penetrant on machined surface.
  - *Eddy current* for sorting, hardness and nodularity (ductile iron) checks.
  - *Leak tests, pressure test* for valve bodies, pump casing; thermal and *acoustic (ring) tests* .
* *Acceptance by classes* (such as ASTM A609, A903, ASME Sec. III NB-2500, MSS SP-54 for valves).
* *Typical castings:* pump casing, valve body, turbine blade, engine block, wheel hubs, gear box.

11. NDE of Welded Constructions

* *Welding defects* come from poor technique, wrong parameters, bad edge preparation or materials. *Weld inspection plan:*

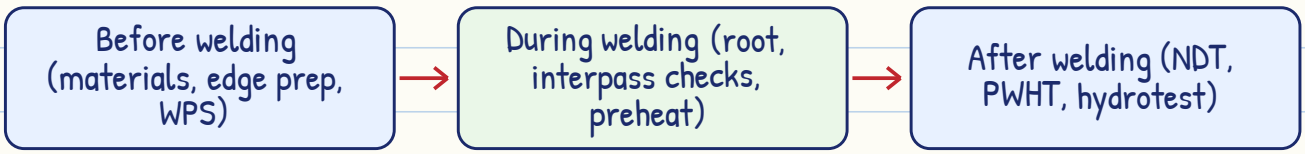


Fig: Three stages of weld inspection

* **Before welding:** check of base metal (UT for laminations), edge preparation, fit-up, WPS / PQR and welder qualification, **consumables** .
* **During welding:** visual of root pass, **MPT / LPT of root and interpass layers** , preheat and interpass temperature control.
* **After welding: visual testing first (VT)** , then **surface NDT** , then **volumetric NDT** , after cooling (hydrogen cracking may appear after 24 - 48 hours; for high strength steels wait before testing).

Typical weld discontinuities found by RT / UT / MT / PT

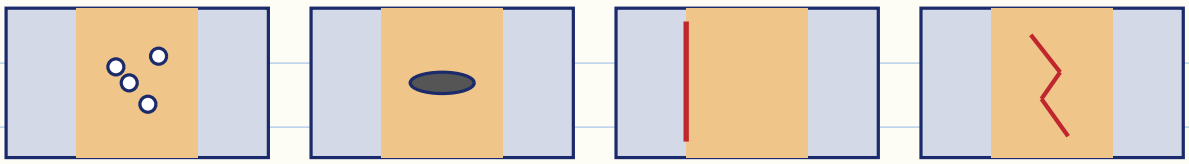


Fig: Common weld defects

Weld Defects and Suitable Methods

Defect	Best detection method
Porosity (round / cluster)	RT, UT (less sensitive)
Slag inclusion	RT, UT
Lack of fusion	UT, RT (if aligned), MPT for surface
Incomplete penetration	RT, UT
Cracks (hot, cold)	UT, MPT / LPT for surface, RT if favourable
Undercut, overlap, spatter	Visual, MPT / LPT
Tungsten inclusion	RT (bright spot)
Distortion, misalignment	Visual and dimensional gauges

Lamellar tearing	UT (normal probe), MPT

* **Radiography** - for butt welds; permanent record; planar flaws may be missed.
* **UT with angle beam, TOFD, phased array** - best for planar flaws in thick welds and in-service inspection; automated for pipelines.
* **MPT / LPT** - for surface and near surface cracks in ferromagnetic / non-magnetic welds.
* **Visual testing** - up to 80% of flaws found; uses weld gauges, magnifier, borescope.
* **ECT / ACFM** - for surface cracks at welds in tubes and offshore joints.
* **Leak testing and hydrotest** - proof of tightness and strength.
* **Acceptance criteria:** ASME Sec. VIII (UW-51 / UW-52), ASME B31.1 / B31.3, API 1104, AWS D1.1 (statically / cyclically loaded structures), ISO 5817 quality levels B / C / D, IS 4853, 2595, 3658, 7307.
* **Records:** NDT procedure, technique sheet, qualified personnel (**ISO 9712 / SNT-TC-1A Level I, II, III**), report with weld map.

Qualification of NDT Personnel

* **SNT-TC-1A, ISO 9712, NAS 410, ISNT (India)** - Level I (performs tests), Level II (sets up and calibrates, interprets, reports), Level III (develops procedures, examines, trains).
* Certification needs training, experience, vision test and examination.

12. Recent Trends in NDE

* **Digital radiography and computed tomography, phased array UT, TOFD, guided wave UT, EMAT, thermography, shearography, laser UT, terahertz imaging, acoustic emission with AI.**
* **Automation and robotics, drone inspection (tanks, chimneys), crawlers, data analytics, machine learning for defect recognition, digital twin and NDE 4.0.**
* **Structural health monitoring (SHM)** - sensors on the structure monitor continuously.
* **Integration with RBI and fitness for service (FFS)** - decisions based on the size of flaw and life.

Summary:

NDE is applied from raw material to in-service in every industry. Railway – UT / ET on rail, axle; Nuclear – remote UT / ECT, strictest; Chemical – corrosion mapping, tubes; Aerospace – ECT, UT, thermography for composites; Automotive – fast MPT, ET, RT in line; Offshore – ROV, ACFM, pigs; Mining – rope MFL; Pressure vessels, castings and welds – RT + UT + MPT / LPT.

Important Questions

* Explain the NDE methods used in the railway and nuclear industries.
* Discuss NDE in aircraft and aerospace industries including composite structures.
* Write NDE applications in the offshore gas, petroleum and coal mining industries.
* How are pressure vessels inspected during fabrication and in service?
* List the defects in castings and welds and the suitable NDT methods for each.